

OSINT Automation Lab

Advanced Osint

M Sameer Malik

OSINT m.sameer.malik113@gmail.com

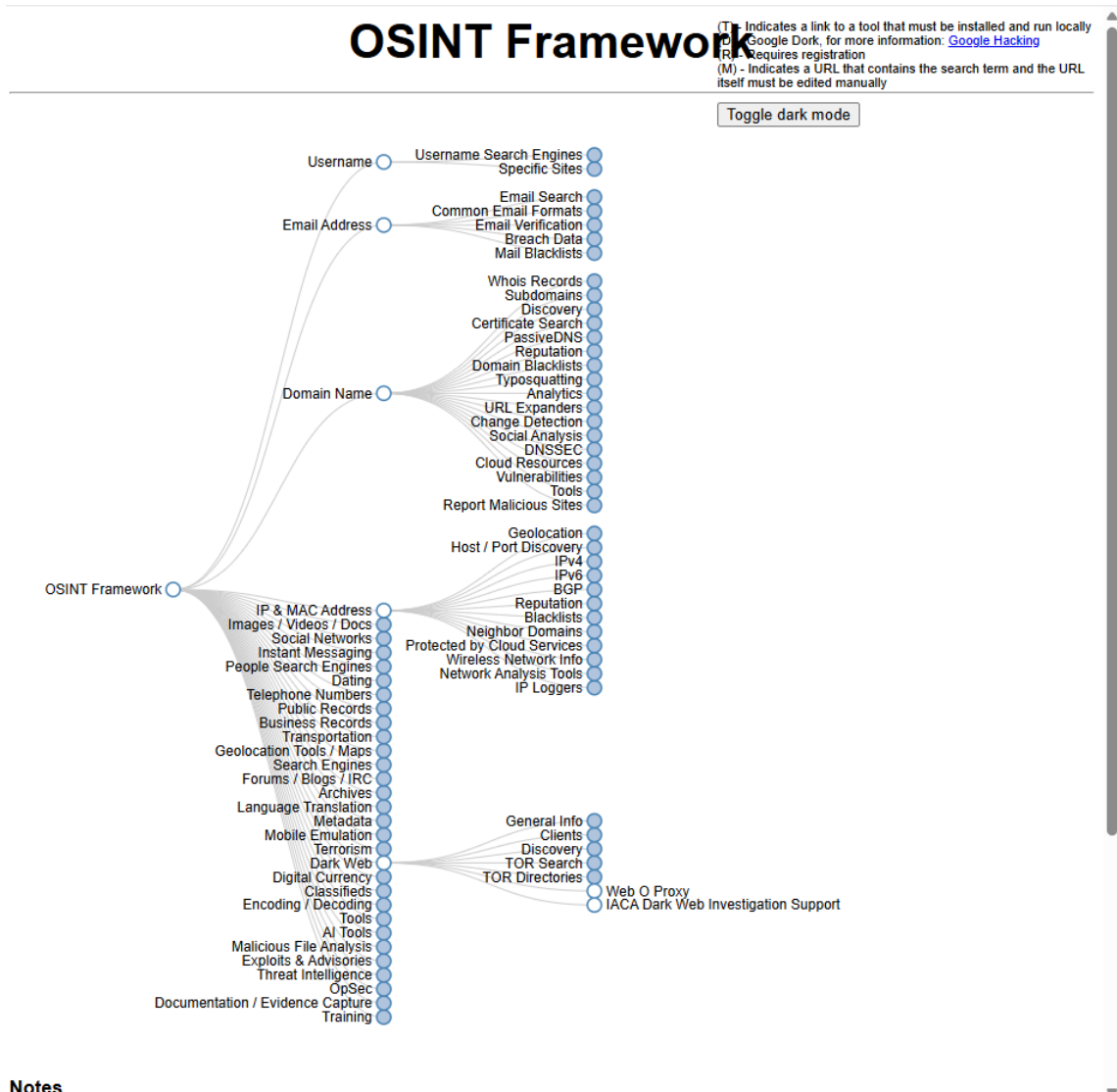
Instructions

Part 1: Examine OSINT Resources

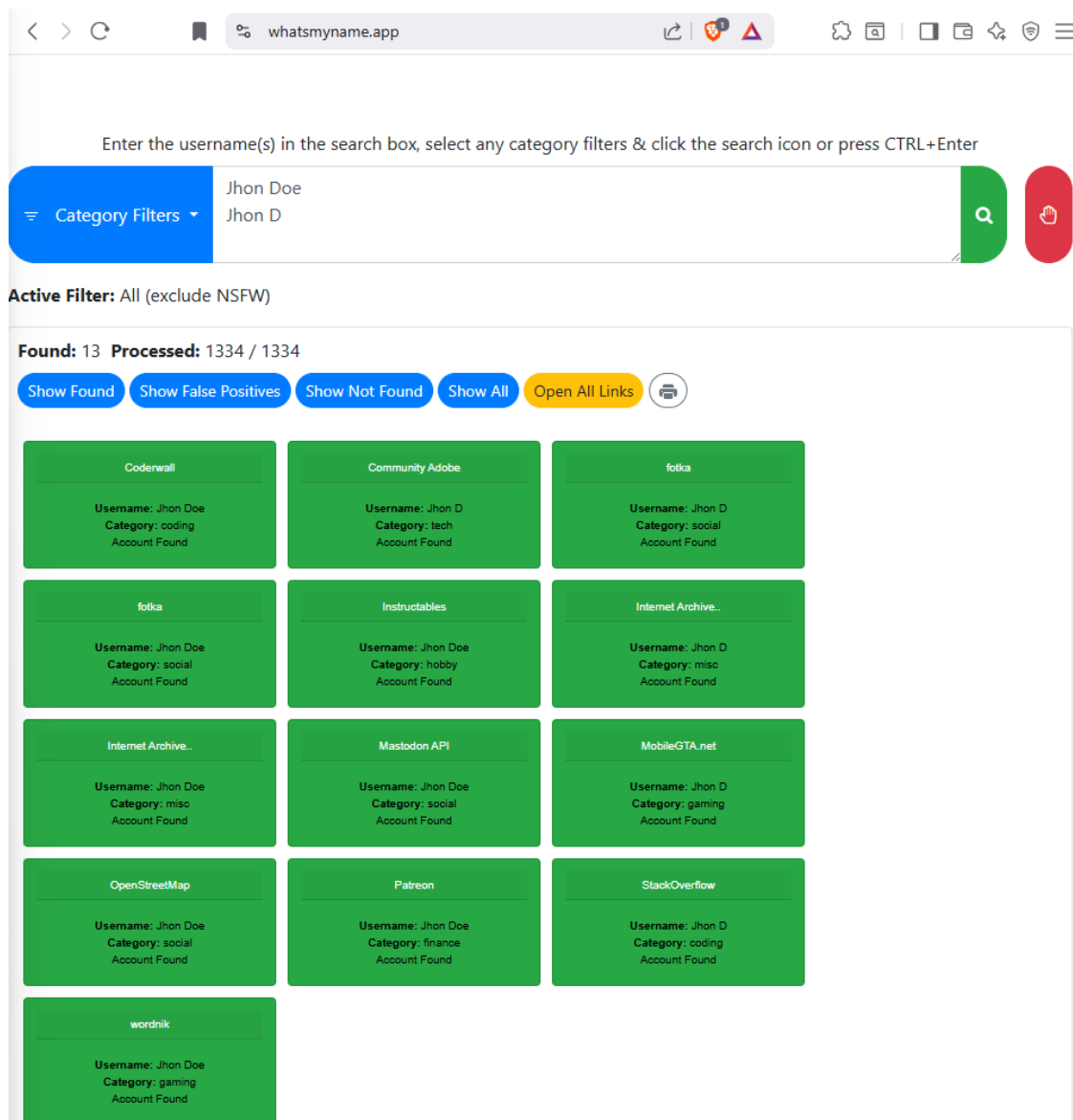
Step 1: Access the OSINT Framework

The OSINT Framework is a useful way to visualize the OSINT tools and resources that are available. Unfortunately, it has become a bit outdated, with some resources no longer available. It is still valuable to help you understand the types of tools available and their uses. In many cases, the links are still good.

- Go to the OSINT Framework site at <https://osintframework.com/>.



- b. Under **Username Search Engines**, click "**WhatsMyName(T)**".



- c. The link takes you to a Git repository for the WhatsMyName project. In the **README.md** content for the tool, the various sites that implement WhatsMyName are listed. Feel free to explore these, but we will click the first link <https://whatsmyname.app/> to visit a free website that implements WhatsMyName.

The parent organization for the site, <https://www.osintcombine.com/>, has several interesting free tools available.

- e. In the search box, type in a few usernames, each on a separate line. Use your own usernames or others that you find. Try searching the internet for **common username wordlist** for other potential search terms. You can filter the results based on the category filters, but for now, just click the green magnifying glass button to start the search.

Category: gaming
Account Found

Filter by Username:
Jhon Doe Jhon D

Show 50 rows▼ Copy CSV PDF

Search:

SITE	USERNAME	CATEGORY	LINK
Coderwall	Jhon Doe	coding	https://coderwall.com/Jhon Doe/
Community Adobe	Jhon D	tech	https://community.adobe.com/t5/forums/searchpage/tab/user?q=Jhon D
fotka	Jhon D	social	https://fotka.com/profil/Jhon D
fotka	Jhon Doe	social	https://fotka.com/profil/Jhon Doe
Instructables	Jhon Doe	hobby	https://www.instructables.com/member/Jhon Doe/
Internet Archive..	Jhon D	misc	https://archive.org/search.php?query=Jhon D
Internet Archive..	Jhon Doe	misc	https://archive.org/search.php?query=Jhon Doe
Mastodon API	Jhon Doe	social	https://mastodon.social/api/v2/search?q=Jhon Doe&type=accounts
MobileGTA.net	Jhon D	gaming	https://www.mobilegta.net/en/user/Jhon D
OpenStreetMap	Jhon Doe	social	https://www.openstreetmap.org/user/Jhon Doe
Patreon	Jhon Doe	finance	https://www.patreon.com/Jhon Doe
StackOverflow	Jhon D	coding	https://stackoverflow.com/users/filter?search=Jhon D
wordnik	Jhon Doe	gaming	https://www.wordnik.com/users/Jhon Doe

In a pentest, you would use another tool, such as **SpiderFoot** (below) to find usernames in email addresses that are associated with a company or domain.

Investigate the results. You can open the links to the accounts either from the green rectangles or the table of results.

WhatsMyName provides a very flexible report of the results. The results table can be sorted by column, and you can export the results as CSV or PDF for reporting purposes. In addition, you can easily filter by username and search within the results. Finally, you get links for the profile pages for the users at many different sites.

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Document Results: these results are document searches with the **first** username in the list used as the search term

[All results](#) [PDF](#) [Spreadsheet](#) [Word Document](#) [PowerPoint](#) [Text Files](#) [Python Code](#) [JavaScript Code](#)

About 2 results (0.09 seconds)

Sort by: [Relevance](#) ▾

Did you mean: "[John Doe](#)"

[Untitled](#)

[akashwebs.com](#) › [invoice](#)

File Format: Microsoft Excel

[jhon doe](#), invoice#, 205092. 9, 14 dummy street area , location,, date, 44907. 10, lorem ipsum , 120xx15xx. 11, NO, ITEM DESCRIPTION, QTY, PRICE, TOTAL. 12. 13.

[Untitled](#)

[www.funcionpublica.gov.co](#) › [admon](#) › [files](#) › [empresas](#) › [archivos](#)

File Format: Microsoft Excel

... [Jhon Doe](#), [jhondoe@hotmail.com](#), iPhone/iPod, Safari, [http://my.clickdesk.com/cdMob?](#)

[id=all_ag9zfmNsaWNrZGVza2NoYXRyDwsSBXVzZXJzGK3pz8kODA&widget_type=ultra_theme ...](#)

[Search for "Jhon Doe" on Google](#)

ENHANCED BY [Google](#)

Part 2: Use SpiderFoot

SpiderFoot is an automated OSINT scanner. It is included with Kali. SpiderFoot queries over 1000 open-information sources and presents the results in an easy-to-use GUI. SpiderFoot can also be run from a console. SpiderFoot seeds its scan with one of the following:

- Domain names
- IP addresses
- Subnet addresses
- Autonomous System Numbers (ASN)
- Email addresses
- Phone numbers
- Personal names

SpiderFoot offers the option of choosing scans based on use case, required data, and by SpiderFoot module. The use cases are:

- All – Get every possible piece of information about the target. This use case can take a very long time to complete.
- Footprint – Understand the target's network perimeter, associated identities and other information that is yielded by extensive web crawling and search engine use.
- Investigate – This is for targets that you suspect of malicious behavior. Footprinting, blacklist lookups, and other sources that report on malicious sites will be returned.
- Passive – This type of scan is used if it is undesirable for the target to suspect that it is being scanned. This is a form of passive OSINT.

Step 1: Start and run SpiderFoot.

In a terminal, enter the following command:

```
(kali㉿kali)-[~]  
└─$ spiderfoot -l 127.0.0.1:5001
```

The command should run without errors. Open a browser and enter the IP address and port for the SpiderFoot GUI. You will see the SpiderFoot interface appear. If this is the first time that SpiderFoot has been opened in this VM, you will see the Scans screen.

```
(kali㉿kali)-[~]  
└─$ sudo apt install spiderfoot  
spiderfoot is already the newest version (4.0-0kali4).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 793  
2025-07-10 07:53:56,190 [INFO] sf : Starting web server at 127.0.0.1:5001 ...  
*****  
Use SpiderFoot by starting your web browser of choice and  
2025-07-10 07:53:56,213 [WARNING] sf :  
*****  
Warning: passwd file contains no passwords. Authentication disabled.  
Please consider adding authentication to protect this instance!  
Refer to https://www.spiderfoot.net/documentation/#security.  
*****  
browse to http://127.0.0.1:5001/  
*****
```

Step 2: Explore SpiderFoot.

```
(kali㉿kali)-[~]  
└─$ spiderfoot -M | grep [search term]
```

```
(kali㉿kali)-[~]  
└─$ spiderfoot -M | grep username  
sfp_keybase          Obtain additional information about domain names and identified usernames.  
sfp_myspace          Gather username and location from MySpace.com profiles.  
sfp_wikipediaedits   Identify edits to Wikipedia articles made from a given IP address or username.  
2025-07-10 08:15:06,507 [INFO] sf : Modules available:
```

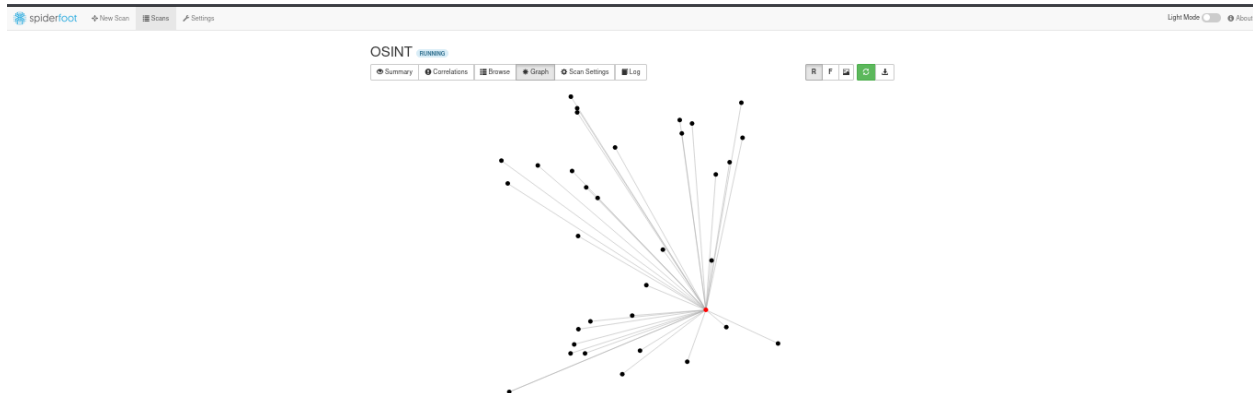
Step 3: Run a SpiderFoot Scan

- Click the **New Scan** tab in the GUI.
- Enter a name for the scan and select a target. In this case, we will use **hackthissite.org**.
- We will scan by use case. Note that you can also scan by the type of information required or by selecting the individual scanner modules that you would like to use. By executing narrower scans, you can learn more about the modules and information that can be gathered.

Note: The **All** use case scan may use active scanning. Unless you have permission to scan the target, you should avoid this setting. To be completely safe, the **Passive** use case should avoid any problems with unauthorized scanning.

- Click the **Run Scan Now** button.

- f. You should see a bar graph appear. The scan statistics will start to increment, and new bars will appear in the graph as new results are obtained. Mouse over the bars for a summary of the findings for that data type.



- b. Note that the terminal prompt changes to indicate that you are working within the Recon-ng framework. Enter **help** to get a sense of the commands that are available.

```
[recon-ng][default] > help
Commands (type [help?] <topic>):
back           Exits the current context
dashboard      Displays a summary of activity
db             Interfaces with the workspace's database
exit           Exits the framework
help           Displays this menu
index          Creates a module index (dev only)
keys           Manages third party resource credentials
marketplace    Interfaces with the module marketplace
modules        Interfaces with installed modules
options        Manages the current context options
pdb            Starts a Python Debugger session (dev only)
script         Records and executes command scripts
shell          Executes shell commands
show           Shows various framework items
snapshots      Manages workspace snapshots
spool          Spools output to a file
workspaces     Manages workspaces
```

- c. Recon-ng uses workspaces to isolate investigations from one another. Workspaces can be created for different parts of a test or different customers for example. Type **workspaces help** to view options for the workspaces command.
- d. Create a workspace named **test** by entering **workspaces create** followed by the workspace name. Note that the prompt has changed to indicate that you are in this workspace.

```
[recon-ng][default] > workspaces create test
[recon-ng][test] > 
```

- e. Type **help** to see the commands that are available within workspaces.

```
[recon-ng][test] > help
Commands (type [help?] <topic>):
back           Exits the current context
dashboard      Displays a summary of activity
db             Interfaces with the workspace's database
exit           Exits the framework
help           Displays this menu
index          Creates a module index (dev only)
keys           Manages third party resource credentials
marketplace    Interfaces with the module marketplace
modules        Interfaces with installed modules
options        Manages the current context options
pdb            Starts a Python Debugger session (dev only)
script         Records and executes command scripts
shell          Executes shell commands
show           Shows various framework items
snapshots      Manages workspace snapshots
spool          Spools output to a file
workspaces     Manages workspaces
```

Step 2: Investigate modules.

Recon-ng is a modular framework. Modules are Python programs with different functions. They are stored in an external marketplace that permits developers to create their own modules and contribute them for use by others.

Return to the Recon-ng prompt. Enter the **modules search** command. This will display the currently installed modules.

```
[recon-ng][test] > modules search
```

```
Recon
```

```
recon/domains-hosts/google_site_web
```

```
[recon-ng][test] > marketplace search
```

Path	Version	Status	Updated	D	K
discovery/info_disclosure/cache_snoop	1.1	not installed	2020-10-13		
discovery/info_disclosure/interesting_files	1.2	not installed	2021-10-04		
exploitation/injection/command_injector	1.0	not installed	2019-06-24		
exploitation/injection/xpath_bruter	1.2	not installed	2019-10-08		
import/csv_file	1.1	not installed	2019-08-09		
import/list	1.1	not installed	2019-06-24		
import/masscan	1.0	not installed	2020-04-07		
import/nmap	1.1	not installed	2020-10-06		
recon/companies-contacts/bing_linkedin_cache	1.0	not installed	2019-06-24	*	*
recon/companies-contacts/censys_email_address	2.1	not installed	2022-01-31	*	*
recon/companies-contacts/pen	1.1	not installed	2019-10-15		
recon/companies-domains/censys_subdomains	2.1	not installed	2022-01-31	*	*
recon/companies-domains/pen	1.1	not installed	2019-10-15		
recon/companies-domains/viewdns_reverse_whois	1.1	not installed	2021-08-24		
recon/companies-domains/whoxy_dns	1.1	not installed	2020-06-17	*	*
recon/companies-multi/censys_org	2.1	not installed	2022-01-31	*	*
recon/companies-multi/censys_tls_subjects	2.1	not installed	2022-01-31	*	*
recon/companies-multi/github_miner	1.1	not installed	2020-05-15	*	*
recon/companies-multi/shodan_org	1.1	not installed	2020-07-01	*	*

Step 3: Investigate the module marketplace.

Recon-ng will not function without modules. In this step, we will install modules from the Recon-ng marketplace. The module marketplace is a GitHub public repository. Search the web for **recon-ng-marketplace** to view the repository. Explore the folders to learn more about the modules.

- In the terminal, view help for the **marketplace** command. Use the **search** option to list all the modules that are currently available.

```
[recon-ng][test][ghdb] > back
```

```
[recon-ng][test] > marketplace list recon
```

```
Interfaces with the module marketplace
```

```
Usage: marketplace <info|install|refresh|remove|search> [...]
```

```
[recon-ng][test] > marketplace search recon
```

```
[*] Searching module index for 'recon' ...
```

Path	Version	Status	Updated	D	K
recon/companies-contacts/bing_linkedin_cache	1.0	installed	2019-06-24	*	*
recon/companies-contacts/censys_email_address	2.1	installed	2022-01-31	*	*
recon/companies-contacts/pen	1.1	installed	2019-10-15		
recon/companies-domains/censys_subdomains	2.1	disabled	2022-01-31	*	*
recon/companies-domains/pen	1.1	installed	2019-10-15		
recon/companies-domains/viewdns_reverse_whois	1.1	installed	2021-08-24		
recon/companies-domains/whoxy_dns	1.1	installed	2020-06-17	*	*
recon/companies-multi/censys_org	2.1	installed	2022-01-31	*	*
recon/companies-multi/censys_tls_subjects	2.1	installed	2022-01-31	*	*
recon/companies-multi/github_miner	1.1	installed	2020-05-15	*	*
recon/companies-multi/shodan_org	1.1	installed	2020-07-01	*	*
recon/companies-multi/whois_miner	1.1	installed	2019-10-15		
recon/contacts-contacts/abc	1.0	installed	2019-10-11	*	*
recon/contacts-contacts/mailtester	1.0	installed	2019-06-24		
recon/contacts-contacts/mangle	1.0	installed	2019-06-24		

Note that the modules are organized by their category and type. This appears as a path prepended to the name of the module. You can filter the output by adding a search term to the marketplace search command. Try a few different search terms that are related to OSINT information to get a sense of the modules that are available.

- d. Instead of passing options at the command line, in Recon-ng you set the options and then enter a simple command to execute the module. Use the **options set source** command to set the only option for this module. Complete the command by specifying the target as **hackthissite.org**.

```
[recon-ng][test][interesting_files] > options set source hackthissite.org
SOURCE => hackthissite.org
[recon-ng][test][interesting_files] > run
[*] http://hackthissite.org:80/robots.txt => 200. 'robots.txt' found!
[*] http://hackthissite.org:80/sitemap.xml => 404
[*] http://hackthissite.org:80/sitemap.xml.gz => 404
[*] http://hackthissite.org:80/crossdomain.xml => 404
[*] http://hackthissite.org:80/phpinfo.php => 404
[*] http://hackthissite.org:80/test.php => 404
[*] http://hackthissite.org:80/elmah.axd => 404
[*] http://hackthissite.org:80/server-status => 200. 'server-status' found but unverified.
[*] http://hackthissite.org:80/jmx-console/ => 404
[*] http://hackthissite.org:80/admin-console/ => 404
[*] http://hackthissite.org:80/web-console/ => 404
[*] 1 interesting files found.
[*] Files downloaded to '/home/kali/.recon-ng/workspaces/test/'
[recon-ng][test][interesting_files] >
```

Now for the practice purposes we will repeat the same steps with another module, I choose the *recon/profiles-profiles/profiler*.

```
[recon-ng][test] > marketplace search recon/profiles-profiles/profiler
[*] Searching module index for 'recon/profiles-profiles/profiler'...

+-----+-----+-----+-----+-----+
| Path | Version | Status | Updated | D | K |
+-----+-----+-----+-----+-----+
| recon/profiles-profiles/profiler | 1.2 | installed | 2023-12-30 | | |
+-----+-----+-----+-----+-----+

D = Has dependencies. See info for details.
K = Requires keys. See info for details.

[recon-ng][test] > marketplace load recon/profiles-profiles/profiler
Interfaces with the module marketplace

Usage: marketplace <info|install|refresh|remove|search> [ ... ]
```

Info command is used to gather general information about a specific module.

```
[recon-ng][test] > marketplace info recon/profiles-profiles/profiler

+-----+-----+
| path | recon/profiles-profiles/profiler |
+-----+-----+
| name | OSINT HUMINT Profile Collector |
+-----+-----+
| author | Micah Hoffman (@WebBreacher), Brendan Burke (@gbinv) |
+-----+-----+
| version | 1.2 |
+-----+-----+
| last_updated | 2023-12-30 |
+-----+-----+
| description | Takes each username from the profiles table and searches a variety of web sites for those users. The list of valid sites comes from the parent project at https://github.com/WebBreacher/WhatsMyName |
+-----+-----+
| required_keys | [] |
+-----+-----+
| dependencies | [] |
+-----+-----+
| files | [] |
+-----+-----+
| status | installed |
+-----+-----+
```

The **load module** command is used to load a specific module into the memory upon which we can operate and execute commands this plays a vital role as it is necessary to load the module in the memory to operate upon it.

```
[recon-ng][test] > modules load recon/profiles-profiles/profiler
[recon-ng][test][profiler] > info

Name: OSINT HUMINT Profile Collector
Author: Micah Hoffman (@WebBreacher), Brendan Burke (@gbinv)
Version: 1.2

Description:
Takes each username from the profiles table and searches a variety of web sites for those users. The
list of valid sites comes from the parent project at https://github.com/WebBreacher/WhatsMyName

Options:


| Name   | Current Value | Required | Description                              |
|--------|---------------|----------|------------------------------------------|
| SOURCE | default       | yes      | source of input (see 'info' for details) |



Source Options:
default      SELECT DISTINCT username FROM profiles WHERE username IS NOT NULL
<string>     string representing a single input
<path>       path to a file containing a list of inputs
query <sql>  database query returning one column of inputs

Comments:
* Note: The global timeout option may need to be increased to support slower sites.
* Warning: Using this module behind a filtering proxy may cause false negatives as some of these
sites may be blocked.

[recon-ng][test][profiler] > options set source hackxor.net
SOURCE => hackxor.net
[recon-ng][test][profiler] > run
```

After running this module we can see that it managed to gather 3 profiles related to the source and each one can viewed separately.

```
[recon-ng][test][profiler] > show profiles
```

rowid	username	resource	url	category	notes	module
1	hackxor.net	chatango.com	https://hackxor.net.chatango.com	social		profiler
2	hackxor.net	GDBrowser	https://gdbrowser.com/u/hackxor.net	gaming		profiler
3	hackxor.net	Sourceforge	https://sourceforge.net/u/hackxor.net/profile	coding		profiler

Find Interesting Files with Recon-ng

In this part of the lab, we will install and use another plugin.

Step 1: Install another module.

- Search the marketplace for a module that will discover interesting files in a domain. The plugin that you use should have no dependencies or key requirements.

```
[recon-ng][test] > marketplace search files
[*] Searching module index for 'files'...
```

Path	Version	Status	Updated	D	K
discovery/info_disclosure/interesting_files	1.2	installed	2021-10-04		
recon/companies-contacts/bing_linkedin_cache	1.0	installed	2019-06-24		*
recon/companies-multi/github_miner	1.1	installed	2020-05-15		*
recon/contacts-profiles/fullcontact	1.1	installed	2019-07-24		*
recon/domains-contacts/metacrawler	1.1	disabled	2019-06-24	*	
recon/profiles-contacts/bing_linkedin_contacts	1.2	installed	2021-08-24		*
recon/profiles-contacts/dev_diver	1.1	installed	2020-05-15		*
recon/profiles-contacts/github_users	1.0	installed	2019-06-24		*
recon/profiles-profiles/namechk	1.0	installed	2019-06-24		*
recon/profiles-profiles/profiler	1.2	installed	2023-12-30		
recon/profiles-profiles/twitter_mentioned	1.0	installed	2019-06-24		*
recon/profiles-profiles/twitter_mentions	1.0	installed	2019-06-24		*
recon/profiles-repositories/github_repos	1.1	installed	2020-05-15		*
recon/repositories-profiles/github_commits	1.0	installed	2019-06-24		*

Set the source option to **hackthissite.org** or another location of your choice. (Please comply with the terms of the course when choosing a domain.) The h4cker.org website is interesting also.

```
[recon-ng][test][interesting_files] > info

Name: Interesting File Finder
Author: Tim Tomes (@alanmaster53), thraprt (thraprt@gmail.com), Jay Turla (@shipcod3), and Mark Jeffery
Version: 1.2

Description:
  Checks hosts for interesting files in predictable locations.

Options:
  Name      Current Value      Required  Description
  ---      -
  CSV_FILE  /home/kali/.recon-ng/data/interesting_files_verify.csv  yes       custom filename map
  DOWNLOAD  True                     yes       download discovered files
  PORT      80                      yes       request port
  PROTOCOL  http                   yes       request protocol
  SOURCE     default                 yes       source of input (see 'info' for details)

Source Options:
  default    SELECT DISTINCT host FROM hosts WHERE host IS NOT NULL
  <string>    string representing a single input
  <path>      path to a file containing a list of inputs
  query <sql> database query returning one column of inputs
```

Now we will set the source to **hackthissite.org** and run the module. As we can see as the module completes it is automatically saved under a directory!

```
[recon-ng][test][interesting_files] > options set source hackthissite.org
SOURCE => hackthissite.org
[recon-ng][test][interesting_files] > run
[*] http://hackthissite.org:80/robots.txt => 200. 'robots.txt' found!
[*] http://hackthissite.org:80/sitemap.xml => 404
[*] http://hackthissite.org:80/sitemap.xml.gz => 404
[*] http://hackthissite.org:80/crossdomain.xml => 404
[*] http://hackthissite.org:80/phpinfo.php => 404
[*] http://hackthissite.org:80/test.php => 404
[*] http://hackthissite.org:80/elmah.axd => 404
[*] http://hackthissite.org:80/server-status => 200. 'server-status' found but unverified.
[*] http://hackthissite.org:80/jmx-console/ => 404
[*] http://hackthissite.org:80/admin-console/ => 404
[*] http://hackthissite.org:80/web-console/ => 404
[*] 1 interesting files found.
[*] Files downloaded to '/home/kali/.recon-ng/workspaces/test/'
[recon-ng][test][interesting_files] >
```

Conclusion:

The following lab demonstrated the use of OSINT tools with a special focus on **OSINT FRAMEWORK**, Spiderfoot and Recon-ng. These automated tools saves a ton of time and empowers to perform deep inspection on the target. Thus leveraging these tools one can boost his efficiency and focus on the work that cant be automated!